0580_w20_qp_42_Q7

7 $y = x^2 + \frac{1}{x}, \ x \neq 0$

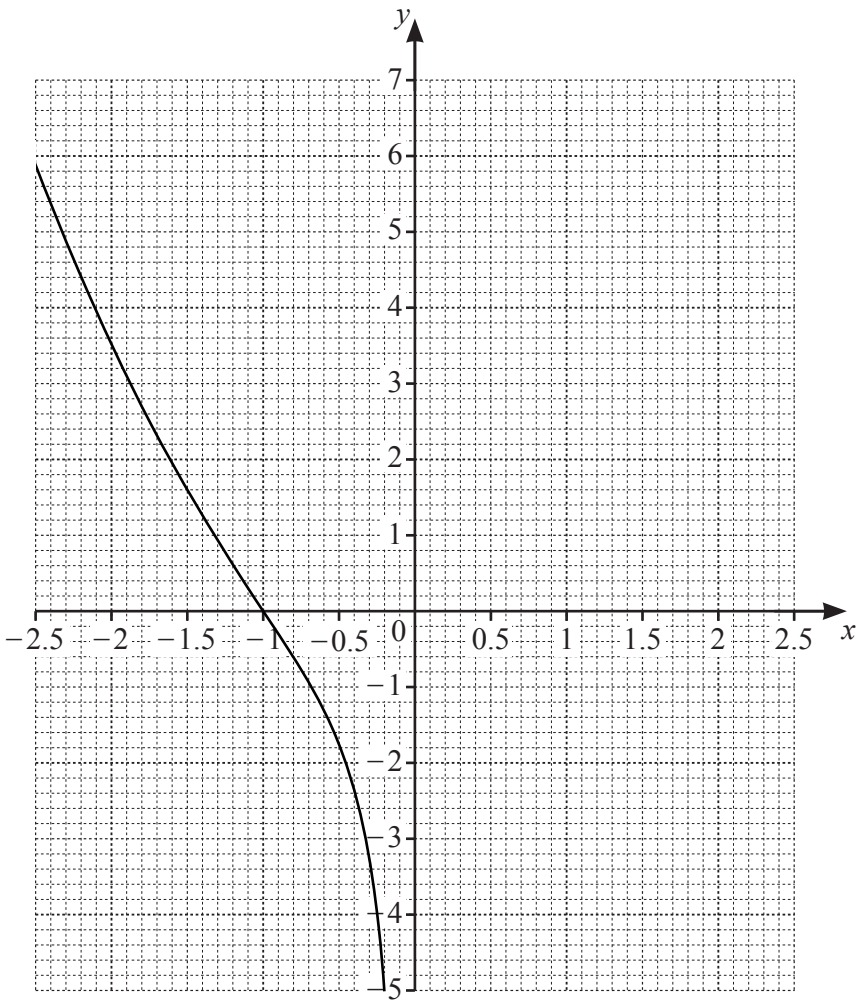
(a) Complete the table.

x	0.2	0.3	0.5	1	1.5	2	2.5
y	5.0	3.4	2.3		2.9		6.7

[2]

(b) On the grid, draw the graph of $y = x^2 + \frac{1}{x}$ for $0.2 \leqslant x \leqslant 2.5$.

The graph of $y = x^2 + \frac{1}{x}$ for $-2.5 \leqslant x \leqslant -0.2$ has been drawn for you.



[4]

0580_w20_qp_42_Q7

(c) By drawing suitable straight lines on the grid, solve the following equations.

(i) $x^2 + \frac{1}{x} = -2$

$x =$ ..... [1]

(ii) $x^2 + \frac{1}{x} + x - 1 = 0$

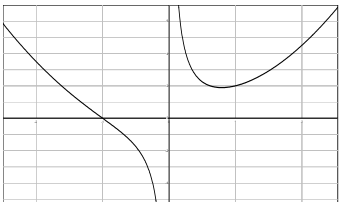
$x =$ ..... [2]

(d) k is an integer and the equation $x^2 + \frac{1}{x} = k$ has three solutions.

Write down a possible value of k.

$k =$ ..... [1]

0580_w20_ms_42_Q7

Question	Answer	Marks	Partial Marks
7(a)	2, 4.5	**2**	**B1** for each
7(b)	Correct graph 	**4**	**B3 FT** for 6 or 7 correct points FT *their* table or **B2 FT** for 4 or 5 correct points FT *their* table or **B1 FT** for 2 or 3 correct points FT *their* table
7(c)(i)	–0.5 to –0.4	**1**	
7(c)(ii)	$y = 1 - x$ ruled **and** –1.9 to –1.75	**2**	**M1** for [y =] $1 - x$ or $\left[x^2 + \frac{1}{x} =\right] 1 - x$ soi or **B1** for –1.9 to –1.75
7(d)	Any integer ≥ 2	**1**	